

Key: Proving a Threshold You Cannot See

84-355 Democracy's Data | Policing | Instructor copy — do not distribute

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All code is given to students. Grade the interpretation.

Structure, as of the rebuild. The brief (`policing-brief.Rmd`) is now a chapter, “A Standard You Cannot Observe”: it is organised by argument under prose headings and carries four numbered D3 figures rather than numbered steps — Figure 1, the same stops under four denominators; Figure 2, what each conditioning step discards; Figure 3, search rate against hit rate for all five groups; Figure 4, the ten-year series. The lab (`policing.Rmd`) is unchanged and still runs Steps 1–10 with the code exposed. **Students should read the brief first. Step numbers below refer to the lab.** The chapter covers the same ground in the same order, so each lab Step has a heading facing it: Steps 1 and 2 = “A traffic stop is a decision, and the file records only the outcome”; Steps 3 and 4 = “The population at risk of being stopped was never counted” (Figure 1); Step 5 = “Take the stop as given”; Step 6 = “Condition once more” (Figure 2); Steps 7 and 8 = “What the searches found” (Figure 3); Step 9 = “Ten years, and the ordering never flips” (Figure 4); Step 10 = “What can be said, and what cannot”, with the infra-marginality objection under “The argument about this method”.

One of three labs in the discrimination unit, alongside redlining and jury selection. They are interchangeable and were built to be run in parallel — split the room, then have each group explain its inference to the others. **The comparison is the payoff:** three different ways of establishing the same kind of claim, with different weaknesses.

This one is the most technically elegant and the most contested. Prepare for a live argument.

Every number below was run against the data file.

The session in one sentence

The comparison everyone reaches for proves nothing, and a second comparison — which never needs to know who was driving — implies that officers searched Black drivers on evidence that would not have been enough for a white driver.

The findings

```
o <- d[order(-d$search_rate), c("race", "search_rate", "hit_rate")]
o$search_rate <- round(o$search_rate, 2); o$hit_rate <- round(o$hit_rate, 1)
o
```

```
#>           race search_rate hit_rate
#> 3         black      15.52      9.2
#> 4       hispanic      9.87     10.2
#> 5         other      3.51     20.3
#> 1         white      3.14     24.2
#> 2 asian/pacific islander    1.81    36.1
```

- Black drivers searched in **15.52%** of stops, white in **3.14%** — **4.9×**.
- Contraband found in **9.2%** of searches of Black drivers, **24.2%** of white.
- Asian/Pacific Islander: searched least (1.81%), highest hit rate (36.1%).
- The pattern is monotone across five groups.

The three things to get said

1. Steps 3-5 are a trap, and students must fall in it first. The stop-share comparison (Step 3) and then the search-rate comparison (Step 5) are what every news story reports, and they genuinely prove nothing — the denominator does not exist. **Do not skip to the good part.** A student who cannot articulate why “Black drivers are searched five times as often” is insufficient has not understood why Step 6 is clever.

2. The outcome test’s move is conditioning (Step 6). Once a search has happened, the question “what did the officer find?” is answerable and the missing denominator is irrelevant. **Make somebody state the logic out loud:** a lower bar means more searches on weaker evidence, and weaker evidence means fewer finds. It is a short argument and it does a great deal of work.

3. Give the objection its due, then weigh it (Step 10). Infra-marginality is real: average hit rates can differ under an identical bar if the underlying distributions differ. Students who like the conclusion will want to wave it away and should not be allowed to. Then make them notice the burden it carries here — it has to explain the Black and Hispanic results *and* the reverse for Asian/Pacific Islander drivers, monotonically, across 905,070 stops.

The line for the board: you cannot see the standard, but you can see what it let through.

Handling the room

This is charged material and the class will contain people with very different priors and, possibly, very different personal experience of traffic stops. Three things that help:

- **Keep it on the inference.** The question on the table is what this table can and cannot establish. That is a question everyone can argue about on equal terms, which a conversation about policing in general is not.

- **Do not let the data settle a question it cannot reach.** It says nothing about any individual officer, and nothing about San Francisco today.
- **Name the alternative explanations yourself, before a student has to.** It changes the temperature of the room when the objections are on the board as part of the lesson rather than as dissent.

Walking the ten steps

Timings assume 45 minutes on this lab. **The whole design is that the answer arrives at Step 7**, after two comparisons have been shown to fail — so do not let the room race ahead.

Step	What it does	Minutes	Watch for
1 What one record is	Establishes that race is the officer's perception and that contraband_found exists only when a search happened	4	Somebody will want to compare this race variable to the census. Head it off here.
2 How many	905,070 stops, 1.6% searched	2	—
3 Stops by race	The table every news story reports	3	Let them draw the wrong conclusion out loud. It is needed for Step 4 to land. The single most important step. Do not rush it.
4 Why it fails	The denominator does not exist and cannot be built	7	
5 Search rates	Conditioning on the stop; 15.52% vs 3.14%	4	
6 The move	State the logic before showing the number	6	
7 Hit rates	9.2% vs 24.2%	3	Make somebody say it out loud , then have the room predict Step 7. The reveal. Let it sit. The Asian/PI point is what makes it hard to explain away. Levels wobble hugely; only the ordering is stable. Say so. Do not let students who like the conclusion wave the objection away.
8 Both at once	Monotone across five groups	5	
9 Every year	Ordering holds 10/10	4	
10 What you can say	Plus infra-marginality	7	

Step 6 is the pivot and deserves the prediction. Ask the room to vote — higher, lower, or the same — before running Step 7. Roughly half a class predicts “higher,” reasoning that more searching means more finding. Having committed to a wrong prediction is what makes the result memorable.

If you are short on time, compress Steps 1-3 and protect Steps 4, 6 and 10. The two comparisons that fail and the objection that survives are the lesson; the numbers are just what makes it concrete.

Option A — Did it change?

Verified output

```
y$hit <- 100 * y$contraband_found / pmax(y$searched, 1)
out <- do.call(rbind, lapply(sort(unique(y$year)), function(yr) {
  b <- y[y$year == yr & y$race == "black", ]; w <- y[y$year == yr & y$race == "white", ]
  if (!nrow(b) || !nrow(w)) return(NULL)
  data.frame(year = yr, black = round(b$hit, 1), white = round(w$hit, 1),
             gap = round(w$hit - b$hit, 1))
}))
out
```

```
#>   year black white gap
#> 1  2007  11.1  26.2 15.1
#> 2  2008   9.9  24.2 14.3
#> 3  2009   8.7  19.3 10.6
#> 4  2010   8.9  27.6 18.8
#> 5  2011   6.0  21.6 15.6
#> 6  2012   5.9  20.7 14.8
#> 7  2013   7.6  21.9 14.3
#> 8  2014   9.4  30.7 21.3
#> 9  2015  13.1  29.8 16.7
#> 10 2016  16.0  28.2 12.2
```

The gap never closes. It runs from 10.6 to 21.3 points across ten years, with no trend.

3/4: Plots it and reports that the disparity persists.

4/4: Notices that the Black hit rate *rises* at the end — 7.6% in 2013 to 13.1% and 16.0% in 2015–16 — which is what you would expect if searches were becoming more selective. Then checks the search rate to see whether fewer searches were being made, rather than asserting a reform from one column.

4/4, best: Observes that the white hit rate moves around just as much (19.3% to 30.7%) with no story attached, so year-to-year wobble is large relative to any trend, and ten points is not many. **Refusing to read a trend into a noisy series is the right answer here**, and it is the same restraint the prediction-market lab asked for.

Option B — Equalise the bar

Verified output

```
b <- d[d$race == "black", ]; w <- d[d$race == "white", ]
hyp <- round(b$stops * (w$searched / w$stops))
c(searches_actual = b$searched,
  searches_at_white_rate = hyp,
  fewer_searches = b$searched - hyp,
  contraband_actual = b$contraband_found,
  contraband_if_hit_rate_held = round(hyp * b$contraband_found / b$searched))
```

```

#>           searches_actual      searches_at_white_rate
#>                23622                4786
#>      fewer_searches      contraband_actual
#>                18836                2183
#> contraband_if_hit_rate_held
#>                442

```

Searching Black drivers at the white rate means **4,786 searches instead of 23,622 — 18,836 fewer, an 80% reduction**. Holding the hit rate constant, finds fall from 2,183 to about 442.

3/4: Does the arithmetic and states the trade — far fewer searches, less contraband found.

4/4: Spots that **holding the hit rate constant is exactly the wrong assumption**, and that the lab’s own argument says so. If the extra searches were the weakest ones, cutting them removes the searches least likely to find anything, so the hit rate should *rise* and the loss should be far smaller than 1,741. A student who says the calculation is an upper bound on the cost has understood the outcome test better than the calculation does.

4/4, best: Asks what the 18,836 searches cost the people searched, and notes that the data records the finds and not the cost — so any cost-benefit framing is measuring one side of the ledger because that is the side somebody wrote down. **That is the course’s recurring point about what gets recorded.**

Option C — The other direction

No verified output beyond the main table; grade the reasoning.

3/4: Applies the test consistently — a high hit rate and a low search rate imply a *higher* bar for Asian and Pacific Islander drivers.

4/4: Is uncomfortable with it and says why. The same logic that indicts the department for Black drivers implies unusually *favourable* treatment here, and students should notice they are more willing to believe the first than the second. **Applying a method only where you like the answer is the failure mode this whole unit is about.**

4/4, best: Notes that “asian/pacific islander” pools very different populations, that the group is a small share of searches, and that the hit rate is therefore less stable — so the honest reading is a wide interval, not a finding. Precision about which conclusions the sample supports is the skill.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code.
- **The move that earns a 4** is distinguishing the two comparisons and explaining why one survives the missing denominator.
- **A 2** is reporting the search-rate gap as the finding, after Steps 3-5 have explained why it is not.
- **Also a 2**, from the other direction: “this proves nothing because correlation is not causation.” The outcome test is not a correlation.
- **Reward anyone who states infra-marginality accurately**, whichever conclusion they draw from it.

Anticipated questions

- “*Doesn’t this just show more crime?*” — No, and this is the question the outcome test exists to answer. More crime in a group would produce a *higher* hit rate, not a lower one. Have this answer ready; it will come up.
- “*Why San Francisco?*” — Because the file is complete and public. Stanford has dozens of jurisdictions; the finding is not unique to this one.
- “*Is this still true?*” — Unknown from here. The data ends in 2016.
- “*Who decides what counts as contraband?*” — Officers, and it varies. A real limitation worth naming: the outcome variable is itself a discretionary judgement, which is the same problem as EAVS’s rejection reasons.

Connections

- **The other two labs in this unit** — redlining (a documented intent, testing whether it still shows) and jury selection (a control group inside the file). **Have the groups compare which inference they trust most.**
- **RPV** — aggregate patterns, individual conclusions, and the gap between them.
- **EAVS** — discretion recorded as a category, and what that hides.